The pattern of prime numbers.

Abstract.

In the field of number theory, the pattern of prime numbers has existed for centuries as an unsolved puzzle, which mathematicians have failed to solve.

However, this paper solved the pattern of prime numbers using an odd number multiplication method, an addition method, and the square root method.

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Introduction.

Ever since natural numbers were discovered, prime numbers have attracted mathematicians due to their properties.

Prime numbers are those numbers that are divisible by 1 and itself.

And around 300 BC, Euclid proved that prime numbers are found infinitely.

And for a long time, mathematicians have been trying to find its pattern. Mathematicians are curious to know how the prime numbers are distributed among the natural numbers.

Although remarkable progress has been made in estimating the number of prime numbers up to a given number N through PNT and prime counting functions,

Also with respect to primality, early investigation of prime numbers through the Miller-Rabin test, and other methods like the square root method and sieve Eratosthenes algorithm show better progress with respect to prime numbers.

However, till now no one has been able to prove how the prime numbers are distributed among the natural numbers, and what pattern do they follow?

Although, despite many efforts, the pattern of prime numbers was not found, the Reimann hypothesis has also failed in finding the pattern of prime numbers.

But through this paper, three methods have been explained that will tell the distribution of prime numbers among natural numbers.
Also, the square root method has been made simpler, and an attempt has been made to explain that to get the prime number, a step-like process has to be followed.

Methodology.

Odd multiplication method for finding prime numbers.

We know that odd numbers are found as follows:

3+10n

5+10n

7+10n

9+10n

1+10n

While prime numbers are also found in the same form

For example,

3+10=13

3+40=43

3+50=53

From the above statement, we will see that prime and odd numbers follow the same pattern.

But odd numbers differ from prime numbers in the following ways:

1. Odd numbers can have more than two factors.
2. Odd numbers are in the form 5+10n, whereas prime numbers are not found in the form 5+10n. Similarly, the prime number is 2.

Mainly odd composite numbers that have more than two factors are found in the following form.

Odd ×Odd.

Od×Prime

Prime ×Prime.

Now we will study the factors of such odd composite numbers whose unit digit numbers are 1, 3, 7, and 9.

Note: Apart from 5 and 2, there will be no prime number whose unit digit numbers are 5 and 2, nor their multiples will be prime numbers.

Now the factors of odd composite numbers having 1, 3, 7, and 9 unit-digit numbers will be as follows:

Factors of odd composite numbers with unit digit number 1.

1×1

9×9

7×3

Factors of odd composite numbers with unit digit number 3.

1×3

9×7

Factors of odd composite numbers with unit digit number 7.

1×7

9×3

Factors of odd composite numbers with unit digit number 9.

1×9

7×7

3×3

Now if these odd numbers are multiplied in increasing order according to the unit-digit number, then we will get prime numbers.

For example, if we want to find the prime numbers having unit digit number 3 up to 100,
So we will multiply odd numbers having unit-digit odd numbers 1, 3, 7, and 9 in the following way.

$3\times11=33$. $7\times9=63$

$3 \times 21 = 63$

$3 \times 31 = 93$

Now we will sort the received numbers in the 3+10n Odd Numbers table as follows.

Table of 3+10n
3
13
23
33
43
53
63
73
83
93

Sorted table 3+10n.

33
63
93

Now the space left in the sorted table of 3+10n is the prime number.
There are 3, 13, 23, 43, 73, and 83 prime numbers from the above table.

Similarly, if we want to find the prime number having unit digit number 7 till 100, then we will again multiply the odd numbers having unit digit numbers 1, 3, 7, and 9 in the following manner:
$7\times11=77$
$3\times9=27$
$3\times19=57$
$3\times29=87$

And if we arrange 7+10n in an ordered table, we will get 7, 17, 37, 47, 67, and 97 prime numbers.

Sorted table 7+10n.

27
57
77
97

Note: One thing to be noted here is that if, by multiplying an odd number up to a given number, more than 1 composite number of the same value is obtained, then we will count the composite number only once in the sorted table.

For example, 9×7=63 and 21×3=63 will be counted once as 63.

Based on the above method, we will find the prime numbers of all unit-digit numbers in the following way.

Table No. 1.

Prime number	1+10n	3+10n	7+10n	9+10n
3	3	3	2	9
13		1	1	
23		1		

Table No. 2.

Prime number	1+10n	3+10n	7+10n	9+10n
7	1	2	1	1
17		1		

Table No. 3.

Prime numbers	1+10n
19	0

Table No. 4.

Prime numbers	1+10n
11	0

Based on the above four tables (1, 2, 3, and 4), we obtain the following composite numbers.

Table 1

3×(11,21,31)

3×(3,13,23)

3×(7,17,27)

3×(9,19,29)

13×(3,7)

23×(3)

33×(3)

Table 2.

7×(11,13,7,9)

17×(3)

Table3

9×(11)

9×0

Table 4.

11×0

In tables no. 3 and 4, no prime number is being multiplied because on multiplication the value of the composite number is above 100, hence only 0 is entered there.
Their only odd composite number 9 is being multiplied.

The odd composite numbers obtained from the above tables, 1, 2, 3, and 4, are given in Table 5 below.
Table no. 5 shows the composite numbers obtained from odd numbers coloured in different colours.
1+10n
3+10n
7+10n
9+10n

Table No. 5.

3+10n	7+10n	9+10n	1+10n
		9	
	27		21
33		39	
		49	
	57		51
63 63		69	
	77		
	87		81

			81
93		99	91

From the above multiplication of odd numbers, we see 23 prime numbers up to 100.
If 2 and 5 are also added to it, there will be a total of 25 prime numbers.
But the thing to be noted here is that 2 and 5 are prime numbers whose use has no role in finding prime numbers till infinity.
Whatever prime numbers we get will be the unit-digit numbers 1, 3, 7, and 9 only.
And by multiplying the odd numbers with these unit-digit numbers, we find out the prime number.
If we find out the prime number from the divisibility rule, then 2 and 5 have no role in it.

Thus, if the total natural numbers are 100%, then there will be 100×40/100=40% odd numbers whose unit digit numbers are 1, 3, 7, and 9.
From here we get a formula to find prime numbers as follows:
We know that the odd numbers having unit digit numbers 1, 3, 7, and 9 are in the form of 3+10, 7+10n, 9+10n, and 1+10n patterns.
And prime numbers remain in the same pattern.
example,
3+10(1)13, a prime number.
3+10(3) 33, odd composite.
Now if the total of odd numbers with given unit digit numbers up to the given number N is 40%, and if the composite numbers with given unit digit numbers are subtracted from these, then we will get all the prime numbers up to the given number N.

Here
N=100
odd numbers with unit digits 1, 3, 7, and 9 = 40

The composite number is obtained by multiplying the odd numbers having unit digit numbers 1, 3, 7, and 9 = 17

Prime numbers with unit-digit numbers1, 3, 7, and 9 up to the given number N =

Odd numbers up to the given number N=40-composite numbers up to the given number N=17=23.

Thus, if 2 and 5 and their multiples are excluded from the counting of odd numbers,

So we will get the following formula.

prime numbers up to given number N = odd numbers up to given number N – composite number up to given number N.

Again, we will get the prime number having unit digit number 3 by the multiple method.

Here N=1000

Table No.6.

Prime numbers.	1+10n	3+10n	7+10n	9+10n
3	33	33	32	32
13	7	7	6	6
23	4	3	3	3
43	2	2	2	2
53	1	1	1	1
73	1	1	1	1
83	1	1	1	1
103				1

113			1	
313		1		

Table No. 7.

Prime numbers.	1+10n	3+10n	7+10n	9+10n
7	14	13	13	14
17	5	5	4	5
37	2	2	2	2
47	2	2	1	2
67	1	2	1	1
97		1	1	1
107		1	1	1
		1	1	1
137		1	1	1
317		1	1	1

Note: In the above tables 6 and 7, when the prime number is multiplied only once until a number N, then we will not multiply further.

Because larger prime numbers have already been multiplied by 3, and 7,

For example, 317×3=351

3×317=351.

Therefore, to avoid repetition, multiplication has to be done once.

Table No. 8.

Prime numbers	1+10n
19	5
29	3
59	1
79	1
89	1

Table No. 9.

Prime numbers	1+10n
11	8
31	3
41	2
61	1
71	1

From the above table, we see that if we divide 1000 by 3 or divide 1000 by 7, we will get the numbers 333.33 and 142.857143.

This means number 3 will be approximately multiplied by 1000/3 = 33 times till 1000, and number 7 will be multiplied by 1000/7 = 14 times.

Below, the multiplication of odd numbers for prime numbers with unit digit number 3 is explained in detail.

On this basis, odd numbers will be multiplied for prime numbers with other unit-digit numbers.
Note: It is important to note here that we will only study the multiplication of odd numbers with unit digit numbers 3×1 and 7×9 to get the prime number with unit digit number 3.
For prime numbers with other unit digits, you can get the composite number from tables No. 6, 7, 8, and 9.

3×(11,21,31,41,51,61,71,81,91,101
111,121,131,141,151,161,171,181,191,201
211,221,231,241,251,261,271,281,291,301
311,321,331)
13×(11,21,31,41,51,61,71)
23×(11,21,31,41)
43×(11,21,)
53×(11)
73×(11)
83×(11)

7× (9,19,29,39,49,59,69,79,89,99
109,119,129,139,)
17× (9,19,29,39,49)
37×(9,19)
47×(9,19)
67×(9)
97×(9)
107×(9)

The composite numbers obtained by colouring with different colours are shown in the table below.

3. 13,7,17,23,43,53,73,83,37,47,67,97,107

Table No. 10.

		203	303	403		603 603	703	803	903 903 903
		213		413	513	623	713	813	913
	123		323	423 423			723		923
33	133		333 333		533	633		833	933
	143	243	343		543			843	943
	153 153	253		453	553		753		
63, 63			363			663 663 663	763		963 963
		273 273 273		473	573			873 873	973
	183			483 483 483	583		783		
93			393	493		693	793	893	993

						693			

We know odd numbers with unit digit numbers 1,7,9 and 3 up to 1000 = 1000×40/100=40.

So,

Odd numbers with unit digit number 3 =400/4=100

Thus we see that the composite numbers with unit digit number 3 are 58 up to 1000.

The Prime number is,

100 – 58 = 42

and as mentioned above, if the same composite number appears more than once in a box, then it is to be considered one composite number.

— To reduce the multiplication of the above odd numbers,

We will exclude multiples of odd numbers from multiplication of odd numbers as follows:

We will focus on the sequence of multiplication of odd numbers up to 1000.

Here the multiples of 3 and 7 are 21, 51, 9, 39, 69, 99, 129, 51, and 49.

So once we multiply 3 and 7, we will not include them again in the multiplication of other prime numbers.

Look.

3×(11,21,31,41,51,61,71,81,91,101

111,121,131,141,151,161,171,181,191,201

211,221,231,241,251,261,271,281,291,301

311,321,331)

13×(11,31,41,61,71)

23×(11,31,41)

43×(11,)

53×(11)

73×(11)

83×(11)

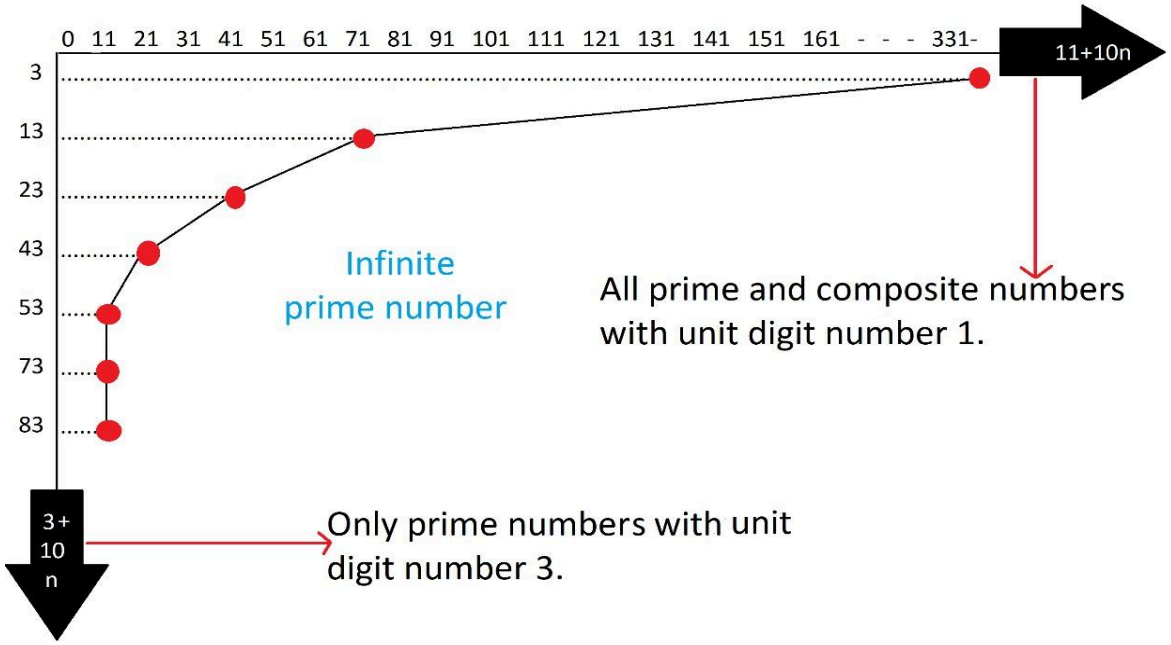
7× (19,29,59,79,89,
109,119,139,)
17× (19,29,49)
37×(19)
47×(19)

In this way, we can avoid the repetition of odd multiplication.

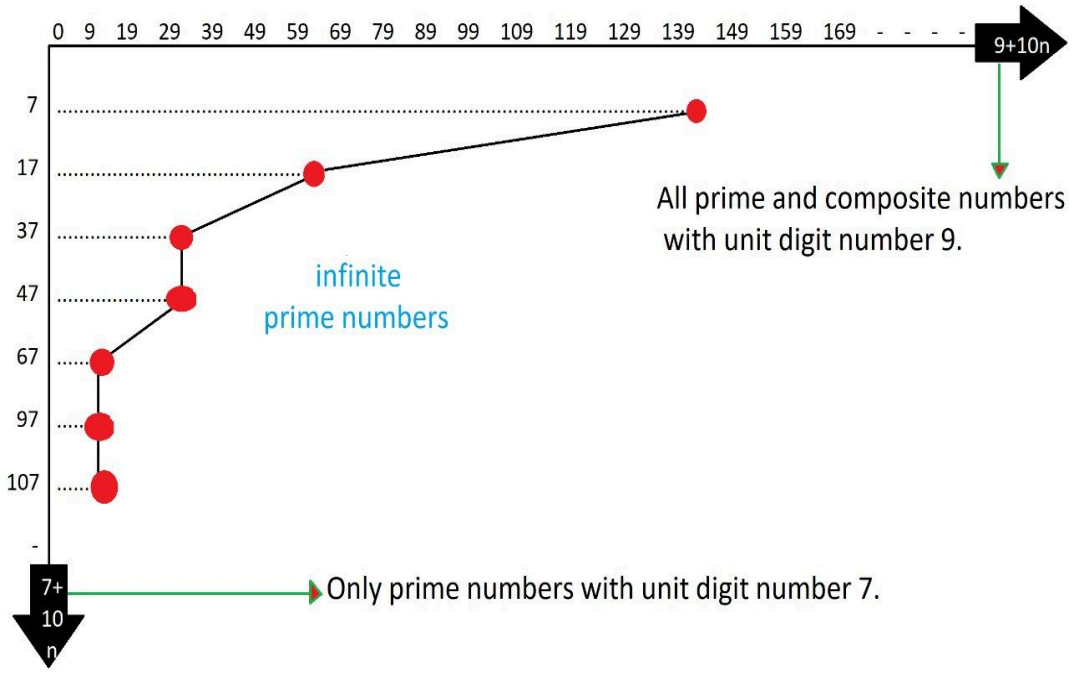
Note: As the number of N sizes increases, multiples of odd numbers will be excluded as 3+10n, 7+10n, 9+10n, and 1+10n.
Along with this, the above exclusion method can also be done in addition to the square root method.

— Given below are graph numbers 1a and 2a, which show the curve of multiplication of odd numbers for a prime number with unit digit number 3.
And along this curve, odd numbers will be multiplied to produce prime numbers.

Graph No.1a.



Graph No.2a.



From the above graphs 1a and 2a, we also come to know that prime numbers are infinite.

— Now we will focus on table no. 4.

Here all prime numbers up to 1000 and composite numbers with unit digit number 3 have been compared.

Table No.11.

N	Total prime numbers up to N	Prime numbers with unit digit number 3	The ratio of all prime numbers and prime numbers having unit digit number 3 to the given number n
100	25	7	25/7=3.57
200	21	5	46/12=3.83
300	16	5	62/17=3.64
400	16	4	75/21=3.71
500	17	3	95/24=3.95
600	14	4	109/28=3.89
700	16	5	125/33=3.78
800	14	3	139/36=3.86

900	**15**	**4**	**154/40=3.85**
1000	**14**	**2**	**168/42=4**

Thus, from the above table, we see that all the prime numbers up to a given number N are approximately 4 times the prime number with unit digit number 3.
Now if the number of prime numbers having one unit digit number up to the given number N is considered as P, then the number of all prime numbers up to the given number N will be approximately **P×4**.

Thus we see that the number of prime numbers having unit digit number 3 up to 1000 is 42.
So if put in the P×4 formula, the number of all prime numbers till 1000 will be =42×4=168.

Thus we see that if we find the number of prime numbers having only one unit digit number up to the given number N, then we can find the number of all prime numbers up to the given number N.
Why is this happening?
You must be seeing from all the above tables that there is not much difference in the number of multiplications of odd numbers till the given number N.
Further, we will also see in the addition method that there is very little difference in the pattern of addition of all the composite numbers

— Till now, we have learned to find prime numbers up to 100 and up to 1000 using odd multiplication methods.
We will pay attention to one more thing here, that is, if we know the prime numbers up to 100, then with the help of these prime numbers, we will get the prime numbers up to 1000 through the multiplication method, and if we know the prime numbers up to 1000. So with the help of these, we will know the prime numbers up to 10000.
That is, from prime numbers up to 10^n, prime numbers up to 10^n+1 will be found.

Through table no. 12 below, it has been said that how many prime numbers are generated from the prime number of 10^n up to 10^n+1?

And how many times are prime numbers up to 10^n producing prime numbers up to 10^n+1?

Table No. 12.

n	**number of prime numbers up to n=x**	**Y=number of prime numbers in the next row/ number of prime numbers in the previous row**	**Prime numbers generated by x up to 10^n+1 = Y×x**
10^1	**4**		
10^2	**25**	**25/4=6.25**	**4×6.25=25**
10^3	**168**	**168/25=6.72**	**25×6.72=168**
10^4	**1229**	**1229/168=7.31547**	**168×7.31547=1229**
10^5	**9592**	**9592/1229=7.8047**	**1229×7.8047=9592**
10^6	**78498**	**78498/9592=8.183**	**9592×8.183=78498**

10^7	**664579**	**664579/78498=8.466**	**78498×8.466=664579**
10^8	**5,761,455**	**5,761,455/664579=8.669**	**664579×8.6695=5761455**
10^9	**50,847,534**	**50,847,534/5761455=8.8254**	**5761455×8.8254=50847534**
10^10	**455,052,511**	**455,052,511/50847534=8.949**	**50847534×8.949=455052511**
10^11	**4,118,054,813**	**4,118,054,813/455052511=9.049**	**455052511×9.049=4118054813**
10^12	**37,607,912,018**	**37,607,912,018/4118054813=9.1324**	**4118054813×9.1324=37604912018**
10^13	**346,065,536,839**	**346,065,536,839/37607912018=9.2019**	**37607912018×9.2019=346065536839**

However, we can estimate the average of prime numbers from 10^n to 10^n+1 from the above table.

But for this, the prime counting function and prime number theorem are already available. Therefore, the number of prime numbers up to a given number N can be accurately predicted by the three methods described in this paper.

The addition method for finding prime numbers.

Through this method, we study the sequence of multiplication of odd numbers and convert them into addition.

We know that odd numbers are found in the following form.

3+10n

7+10n

9+10n

1+10n

And these will be multiplied in the following order.

Multiplication of odd numbers having unit digits 1 and 3 to get a prime number with unit digit number 3.

Table No. 13.

3×11=33	3×21=63	3×33=93	Horizontal addition 3+30n.
13×11=143	13×21=273	13×31=403	
23×11=253	23×21=483	23×31=713	
33+110n Vertical addition.	33+30×1+210n	33+30×2+310n	Vertical addition. 3+30n+110+100n

Multiplication of odd numbers having unit digit numbers 7 and 9 to get a prime number with unit digit number 3.

Table No. 14.

7×9=63	7×19=133	7×29=203	Horizontal addition 63+70n.
17×9=153	17×19=323	17×29=493	
27×9=243	27×19=513	27×29=783	
63+90n Vertical addition.	63+70×1+190n	63+70×2+290n	Vertical addition. 63+70n+90+100n

Multiplication of odd numbers having unit digit number 7 and 1 to get a prime number with unit digit number 7.

Table No. 15.

7×11=77	7×21=147	7×31=9 217	Horizontal addition 7+70n.
17×11=187	17×21=357	17×31=527	
27×11=297	23 7×21=567	27×31=837	
77+110n Vertical addition.	77+70×1+210n	77+70×2+310n	Vertical addition. 77+70n+110+100n

Multiplication of odd numbers having unit digit numbers 3 and 9 to get a prime number with unit digit number 7.

Table No. 16.

3×9=27	13×9=117	23×9=9 207	Horizontal addition 27+90n.
13×9=111	13×19=247	23×19=437	
23×9=207	13×29=377	23×29=667	

27+90n Vertical addition.	27+90×1+1900n	27+90×2+290n	Vertical addition. 27+90n+90+100n

Multiplication of odd numbers having unit digit numbers 3 and 3 to get a prime number with unit digit number 9.

Table No.17.

3×3=9	3×13=39	3×23=69	Horizontal addition 9+30n.
13×3=39	13×13=169	13×23=299	
23×3=69	23×13=299	23×23=529	
9+30n Vertical addition.	9+30×1+130n	9+30×2+230n	Vertical addition. 9+30n+130+100n

Multiplication of odd numbers having unit digit numbers 7 and 7 to get a prime number with unit digit number 9.

Table No. 18.

7×7=49	7×17=119	7×27=189	Horizontal addition 49+70n.
17×7=119	17×17=289	17×27=459	
27×7=189	27×17=459	27×27=729	
49+70n Vertical addition.	49+70×1+170n	49+70×2+270n	Vertical addition. 49+70n+70+100n

Multiplication of odd numbers having unit digit number 9 and 1 to get a prime number with unit digit number 9.

Table No. 19.

9×11=99	9×21=189	9×31=279	Horizontal addition 9+90n.

19×11=209	19×21=399	19×31=589	
29×11=319	29×21=609	29×31=899	
99+110n Vertical addition.	99+90×1+210n	99+90×2+310n	Vertical addition. 9+90n+110+100n

Multiplication of odd numbers having unit digit numbers 1 and 1 to get a prime number with unit digit number 1.

Table No. 20.

11×11=121	11×21=231	11×31=341	Horizontal addition 11+110n.
21×11=231	21×21=441	21×31=651	
31×11=341	31×21=651	31×31=961	
121+110n Vertical addition.	121+110×1+210n	121+110×2+310n	Vertical addition. 121+110n+110+100n

Multiplication of odd numbers having unit digit numbers 3 and 7 to get a prime number with unit digit number 1.

Table.No. 21.

3×7=21	3×17=51	3×27=81	Horizontal addition 21+30n.
13×7=91	13×17=221	13×27=351	
23×7=161	23×17=391	23×27=621	
21+70n Vertical addition.	33+30×1+170n	21+30×2+270n	Vertical addition. 21+30n+70+100n

Multiplication of odd numbers having unit digit numbers 9 and 9 to get a prime number with unit digit number 1.

Table.No. 22.

9×9=81	9×19=171	9×29=261	Horizontal addition 81+90n.
19×9=171	19×19=361	19×29=551	
29×9=261	29×19=551	29×29=841	
81+90n Vertical addition.	81+90×1+190n	81+90×2+290n	Vertical addition. 81+90n+90+100n

The sequence of multiplication and addition of odd numbers given in tables 23 to 32 will mainly be used in the square root method.

Multiplication of odd numbers having unit digits 1 and 3 to get a prime number with unit digit number 3.

Table No. 23.

11×3=33	11×13=143	11×23=253	33+110n Horizontal addition.
11×13=143	11×23=253	11×33=363	
33+110n Vertical addition.	33+110×1+110n	33+110×2+110n	33+110n+110n Vertical addition

Multiplication of odd numbers having unit digit numbers 9 and 7 to get a prime number with unit digit number 3.

Table No. 24.

9×7=63	9×17=153	9×27=243	63+70n Horizontal addition.
19×7=133	19×17=323	19×27=513	

63+70n Vertical addition.	63+70×1+170n	63+70×2+270n	63+70n+170+100n Vertical addition.

Multiplication of odd numbers having unit digit numbers 1 and 7 to get a prime number with unit digit number 7.

Table No. 25.

11×7=77	11××17=187	11×27=297	77+110n Horizontal addition.
21×7=147	21×17=357	21×27=567	
77+70n Vertical addition.	77110×1=170n	77+110×2=270n	77+110n+170+100n Vertical addition.

Multiplication of odd numbers having unit digit numbers 9 and 3 to get a prime number with unit digit number 3.

Table No. 26

9×3=27	9×13=117	9×23=207	27+90n Horizontal addition.

19×3=57	19×13=247	19×23=437	
27+30n Vertical addition.	27+90×1+130n	27+90×2+230n	27+90n+130+100n Vertical addition.

Multiplication of odd numbers having unit digit numbers 3 and 3 to get a prime number with unit digit number 9.

Table No. 27.

3×3=9	3×13=39	3×23=69	Horizontal addition 9+30n.
13×3=39	13×13=169	13×23=299	
23×3=69	23×13=299	23×23=529	
9+30n Vertical addition.	9+30×1+130n	9+30×2+230n	Vertical addition. 9+30n+130+100n

Multiplication of odd numbers having unit digit numbers 7 and 7 to get a prime number with unit digit number 9.

Table No. 28.

7×7=49	7×17=119	7×27=189	Horizontal addition 49+70n.
17×7=119	17×17=289	17×27=459	
27×7=189	27×17=459	27×27=729	
49+70n Vertical addition.	49+70×1+170n	49+70×2+270n	Vertical addition. 49+70n+70+100n

Multiplication of odd numbers having unit digits 1 and 9 to get a prime number with unit digit number 9.

Table No.29.

11×9=99	11×19=209	11×29=319	99+110n Horizontal addition.
21×9=189	21×19=399	21×19=609	
99+90n Vertical addition.	99+110×1+190n	99+110×2+290n	99+110n+190+100n Vertical addition.

Multiplication of odd numbers having unit digit numbers 7 and 3 to get a prime number with unit digit number 1.

Table No.30.

7×3=21	7×13=91	7×23=161	21+70n Horizontal addition.
7×13=91	7×23=161	7×33=231	
21+70n Vertical addition.	21+70×1+70n	21+70×2+70n	21+70n+70n Vertical addition.

Multiplication of odd numbers having unit digit numbers 9 and 9 to get a prime number with unit digit number 1.

Table No. 31.

9×9=81	9×19=171	9×29=261	Horizontal addition 81+90n.
19×9=171	19×19=361	19×29=551	
29×9=261	29×19=551	29×29=841	
81+90n Vertical addition.	81+90×1+190n	81+90×2+290n	Vertical addition. 81+90n+90+100n

Multiplication of odd numbers having unit digits 1 and 2 to get a prime number with unit digit number 1.

Table No.32.

11×11=121	11×21=231	11×31=341	Horizontal addition 11+110n.
21×11=231	21×21=441	21×31=651	
31×11=341	31×21=651	31×31=961	
121+110n Vertical addition. As	121+110×1+210n	121+110×2+310n	Vertical addition. 121+110n+110+100n

Here we will find the prime numbers through the sequence of addition and multiplication of unit digit numbers 1, 3, 7, and 9 to get the prime number with unit digit number 3.

We know The horizontal addition of multiplication of odd numbers having unit digit number 3×1 and 9×7 will be obtained as follows.

3+30n

63+70n

63+90n

And the vertical addition of multiplication of odd numbers having unit digit numbers 3 and 1 and 9 and 7 will be obtained as follows:.

33+110n

33+30×1+110n

$33+30\times2+210n$

$33+30\times3+310n$

$33+30\times n+110+100n$

The vertical addition of multiplication of odd numbers having unit digit numbers 7 and 9 will be obtained as follows:

$63+70\times1+1$follows:

$63+70\times2+290n$

$63+70\times3+390n$

$63+70n+190+100n$.

Now if we want to know the prime numbers up to 523, then they will be the addition method in the following way.

Here N=523

1.3+30(17)

n=523/30=17

2. 33+110(4)

n=523/110=4

3. 33+30×1+210(2)

n=523/210=2

4. 33+30×2+310(1)

523/310=1

1.63+70n

n=523/70=7

2.63+90(5)

m=523/90=5

3.63+70×1+190(2)

523/170=3

4.63+70×2+290(1)

Table No. 33.

3+10n Row, Ex.3,13,23.	3	13	23	

33	43	53	63 63	73
83	93	103	113	123
133	143	153 153	163	173
183	193	203	213	223
233	243 243	253	263	273 273 273
283	293	303	313	323
333 333	343	353	363 363	373
383	393	403	413	423 423

433	**443**	**453** **453**	**463**	**473**
483 **483** **483**	**493**	**503**	**513** **513** **513**	**523**

From the above table no. 33, we see that 26 numbers colored in **pink** are prime numbers.

Square root method to find prime numbers.

Under this method, we find the prime number by taking the square root of the given number N.

For example,

N=1000

√1000=31

Here 31 means that all the odd prime numbers up to 1000 will be divided by the prime numbers from 3 to 30.

And if the odd numbers up to the given number N are not divisible, then they will be prime numbers.

Here, around 3 to 30 will be those odd prime numbers whose unit digit numbers are 1, 3, 7, and 9.

Another thing to be noted here is that if we take the square root of a given odd number N, then we should know which unit digit number will be the factors of the given odd number.

For example

N=913

So we saw in the multiplication method that the factors of the odd number having unit digit number 3 will be the odd numbers having unit digit numbers 1, 3, 7, and 9.

Now √913=31

So this means that out of all the 913 factors, there will be one such factor, which will be around 3 to 30.

From the above example

√913=31

The factors of 913 are 11×83.

So we saw that even though 83 is above 30, 11 is below 30.

Here N=1000, then √1000=31 means the following prime+odd numbers will be around 3 to 30,

3,13,23

7,17,27

9,19,29

11,21,31.

Note: 31/10=3.1 means 3 numbers of any one unit digit number out of 4 unit digit numbers 1, 3, 7, and 9.

So the number of all unit-digit numbers 1, 3, 7, and 9 will be 3×4=12.

But among these 12 numbers, odd numbers are having 1, 3, 7, and 9 unit digit numbers, which are 100%.

But the number of prime numbers will always be less than the square root of the given number N.

We see this again!

For example,

N=100

√100=10

10/10=1

This means that any odd number up to 100 will be divided by less than 4 prime numbers.

Like 91 is divisible by 7 and 13, but among the prime numbers having unit-digit numbers 1, 7, 9, and 3, one will be the prime number having less than 10.

For example-

3×(11,21,31,)=33,63,93

3×(3,13,23,33)=9,39,69,99

3×(7,17,27)=21,51,81

3×(9,19,29)=27,57,87

7×(11)=77,

7×(9)=63

7×(3,13)=21,91 ,7×(7)=49

— Now prime numbers up to 10 crores can be found from the square of 31^2.

And prime numbers up to 1e12 will be found from the square of 961^2=, and this process will go on to infinity.

Given below is a table to find the prime numbers from the square root of the number N up to the N number.

Table No. 34.

N	The number of all composite numbers and prime numbers up to the square root of a given number N, =N/10×4=12	Finding Prime Numbers from Square Root to Number N	N^2	Finding Prime Numbers from Square Root to Number N
31	31/10=3.1 4×3.1=12	1000	31^2=961	100000000
961	961/10=384 384×4=1536	100000000	961^2=923521	1000000000000
923561	923561/10=9235 2×4=369408	8.52891037e11	923561^2=8.528 91037e11 Approx 1e12.	1e24
1e12	1e12/10×4e11	1e24	1e24	1e48
1e24	1e24/10×4e23	1e48	1e48	1e96
1e48	1e48/10×4e47	1e96	1e96	1e192
1e96	1e96/10×4e95	1e192	1e192	1e384
1e192	1e192/10×4e191	1e384	1e384	1e768
1e384	1e384/10×4e383	1e768	1e768	1e1536

1e768	1e768/10×4=4e767	1e1536	1e1536	1e3072

Through the above table, we understood that if we know the prime number X up to the square root of a given number **n**, then through X^2 we can find the prime number up to n^2.
And we can say,
√n=X
Then,
X^2=n^2

Till now, we have known prime numbers up to 1e30 through computer calculations. Now if we know prime numbers up to 1e30, then with its help we can easily know prime numbers up to 1e60, and if we know prime numbers up to 1e60, then with its help we can know prime numbers up to 1e120, and with its help we can know prime numbers up to 1e140.
Thus, if we proceed stepwise, we can easily find prime numbers close to infinity.

Now we can find the prime number by combining the square root method with the addition method and multiplication method in the following way.

Square root method + odd multiplication method.

√100=10
Now, the prime numbers below 10 are 3 and 7.
Now, this prime number will be multiplied by odd numbers below 100 to form a composite number as follows:

3×(11,21,31,)=33,63,93
3×(3,13,23,33)=9,39,69,99

3×(7,17,27)=21,51,81

3×(9,19,29)=27,57,87

7×11=77,

7×9=63

7×3,13=21,91

7×7=49

Based on the above multiplication, we get 18 composite numbers.

Whereas 63 and 21 have come twice.

Thus total composite numbers = 18-2 = 16.

Now by subtracting the above composite number from the odd number, we will get the prime number

All odd numbers up to 100 = 100×40/100 = 40

Since 1 is neither an odd nor a prime number, the total number of odd numbers up to 100 is 39.

Thus ,

39-16=23

Thus we see that there are 23 prime numbers up to 100 that are related to the unit digit numbers 1, 3, 7, and 9.

If 2 and 5 are also added to it, there will be 23 + 2 = 25 prime numbers.

Square root method + addition method.

√100=10

Now, the prime numbers below 10 are 3 and 7.

Now, through the addition sequence of these prime numbers, we will obtain composite numbers as follows:.

33+30n

We will obtain **n** in the following way.

100/30=approx 3,

So,

33+30(2)=33,63,93

Here n = 3 would mean that 30 will be added thrice. But here, by adding 30 thrice, the order of addition of the composite number is going beyond 100; hence, we will add 30 only twice. Similarly, you can get the value of **n** from other numbers.

63+70n=63

77+70n=77

27+90n=27,

9+30n=9,39,69

49+70n=49

99+90n=99

21+70n=21,91

21+30n=21,51,81

81+90n=81

27+30n=27,57,87

Note: The addition method mentioned above is the sequence addition of composite numbers. So for example, if it is 77+70n, then the numbers obtained under 77 and 77+70n will be composite numbers.

Because composite number 77+70=147 is above 100

Therefore, we will count only 77.

Based on the above addition method, we get 20 composite numbers,

Whereas 63, 21, 21, 27, and 81 have come twice.

Thus total composite numbers = 20-4 = 16.

Now by subtracting the above composite number from the odd number, we will get the prime number 39-16=23. If 2 and 5 are also added to it, there will be 23 + 2 = 25 prime numbers.

Result.

After presenting all three methods, we get the following result:

1. The number of prime numbers up to a given number N is obtained by subtracting the number of composite numbers up to the given number N from all the odd numbers up to the given number N.

2. If a prime number with one unit digit number is known, then by putting it in the p×4 formula, there will be the least possibility of error in knowing the number of all prime numbers.

3. Prime numbers follow the same pattern as multiplication or addition of odd numbers.

4. Prime numbers up to n^2 can be found by squaring the square root of the given number N.

5. To find prime numbers up to any number N, it would be better to follow a stepwise process.

Conclusion.

Through this paper, we have learned the distribution of prime numbers through the odd multiplication method, addition, and square root method, which follow the pattern of multiplication and addition of odd numbers.

Apart from this, through this paper we learned that to know the prime number with infinite precision, there is a need to proceed stepwise.

In this way, the problem that has been going on since ancient times has been solved through this paper.

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